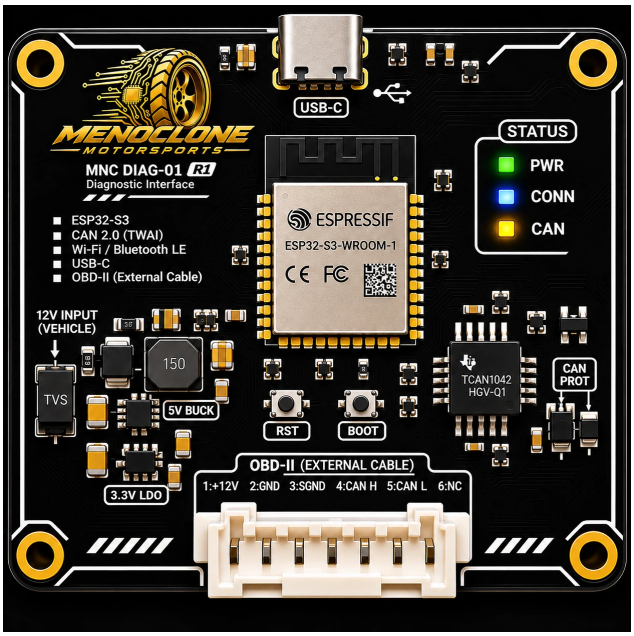
MNC DIAG-01

CAN based vehicle diagnostics development concept

STATUS: IN DEVELOPMENT | PRELIMINARY SPECIFICATIONS



Overview

An automotive diagnostic interface concept exploring supported OBD-II requests, vehicle telemetry, CAN communication and software dashboards.

Proposed technical specifications

Microcontroller	ESP32-S3-WROOM-1 (proposed)
CAN controller	ESP32-S3 TWAI CAN 2.0 controller (proposed)
CAN physical layer	External CAN transceiver (proposed)

Wireless	Wi-Fi and Bluetooth LE (proposed)
Host interface	USB-C (proposed)
Vehicle interface	External OBD-II cable (proposed)
Input power	Nominal 12 V vehicle supply with protection and regulation (proposed)
Indicators	Power, connection and CAN activity LEDs (proposed)

Intended applications

Supported CAN based OBD-II diagnostics, trouble code reading, selected live parameter monitoring, controlled bench testing, educational CAN experiments and prototype dashboards.

Engineering and safety notes

Illustration and connector labels are not a verified wiring diagram. OBD-II standard pin assignments must be confirmed in the electrical design. Automotive surge, reverse polarity, ESD, CAN termination, electrical isolation considerations and firmware safety must be engineered and tested before connecting to any vehicle. Compatibility is not universal.